

Choose your robot's next job

Name: _____

Date: _____ Period: _____

Choose one of five builds, test it and explain how you improved it.

Choose ONE build

1. Parcel delivery robot - full module: 5-8

Carry a paper parcel to a red-line delivery space.

2. Drawing robot - full module: 9-12

Draw a short line on a protected paper surface.

3. Paper sweeper - full module: 13-16

Push three light paper pieces into a collection area.

4. Ball-pushing robot - full module: 17-20

Guide one light paper ball toward a goal.

5. Rescue carrier - full module: 21-24

Tow a light paper figure on a small rescue sled.

Your teacher will give you the pages for one build. Class 1: choose and build. Class 2: test, improve and share. The page numbers above refer to the complete module workbook.

Plan before you build

Our chosen job:

Why we chose it:

LOOK AT THE IDEA



Keep your notes. Change one thing. Try again.

Words to know

Attachment: A part you add to the robot for a new job.

Evidence: Measurements and notes that show what happened.

Handover: Directions that help someone else use your robot.

Draw the attachment and label where it joins the base.

- ☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

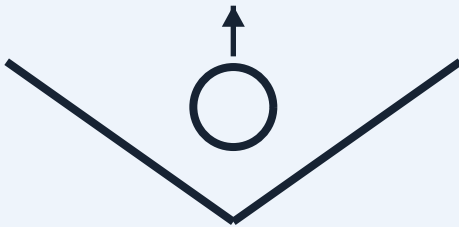
Choice 4: Ball-pushing robot

Name: _____

Date: _____ Period: _____

Guide one light paper ball toward a goal.

LOOK AT THE IDEA



Top view: a shallow V

The opening faces forward.

Leave room for the ball to roll.

Gather your materials

- The shared checked base
- Thin card about 14 by 5 cm
- One crumpled paper ball about 3 cm wide
- Removable tape and ruler

Use the same teacher-checked base. The drawing explains the attachment; it is not a full-size cutting template.

Build your attachment

- 1 Fold the card lengthwise so it has a 3 cm upright wall and a 2 cm mounting flap. Fold it gently across the middle to make a shallow V.
 - 2 Make one light paper ball about 3 cm wide. Put it in the V opening. Leave enough room for it to roll freely; do not make a tight clamp.
 - 3 Turn the robot off. Attach the two mounting flaps to teacher-approved fixed front points. Keep the V open at the front. Leave the wheels clear and keep the card just above the floor.
 - 4 Keep the sensor clear even though this task uses a timed drive. Mark a wide goal across a short straight lane. Place the ball at the same start mark each time.
 - 5 Use the short program below. The robot pushes the ball; it does not grip it. Stop if the ball rolls toward people or outside the clear test area.
 - 6 Repeat three times. If the ball escapes sideways, adjust the V angle a little. Keep the same ball and code while comparing.
- ☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.